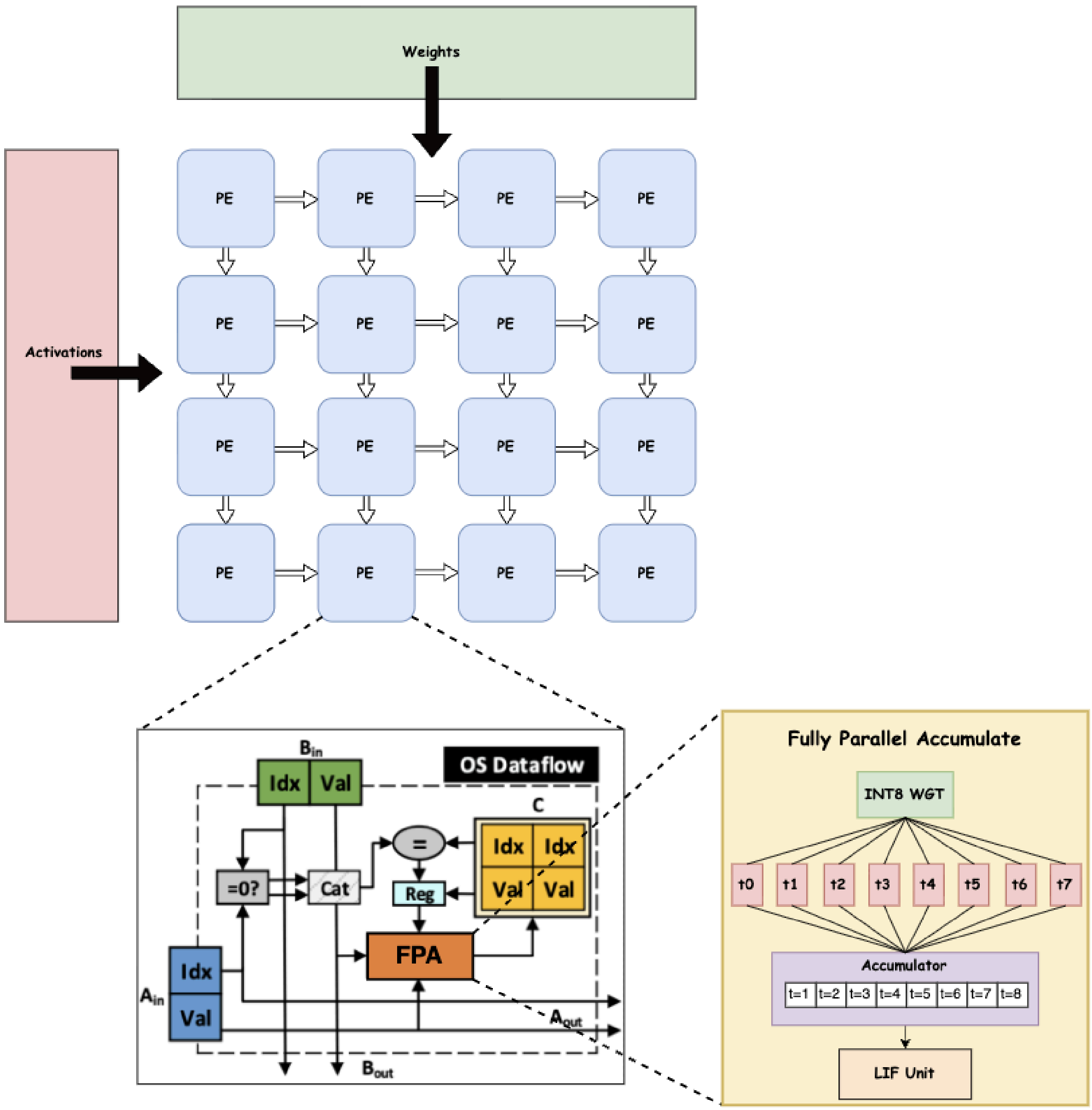
Architecture



1.Compression Ratio

define A as an $M \times K$-scale matrix with sparsity of S, while NNZ is the amount of nonzero elements in A and M_b is the storage overhead of A. The scale of systolic arrays are $K \times K$.

Compression Ratio. A^* is an $R \times K$ matrix with sparsity of S_c and storage cost M_c. We can obtain the formula as follows:

$$NNZ = M \cdot K \cdot (1 - S) = R \cdot K \cdot (1 - S_c) \leq R \cdot K \tag{1}$$

$$M_b = M \cdot K \cdot dataWidth \tag{2}$$

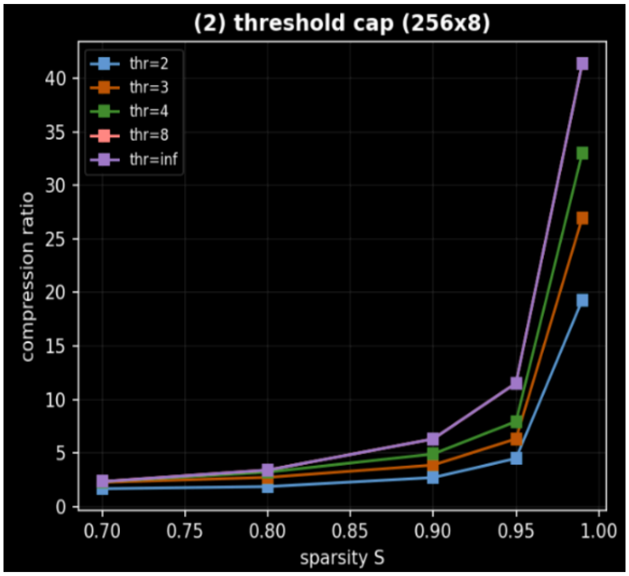
$$M_c = R \cdot K \cdot (dataWidth + indexWidth) \tag{3}$$

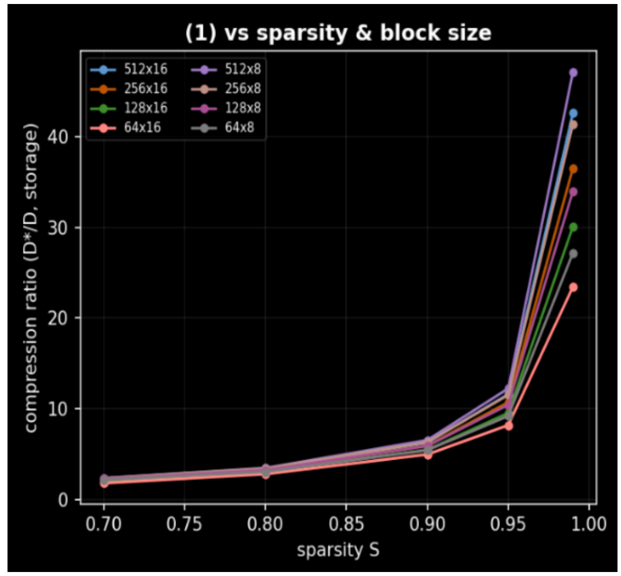
$$indexWidth = \log_2 M \tag{4}$$

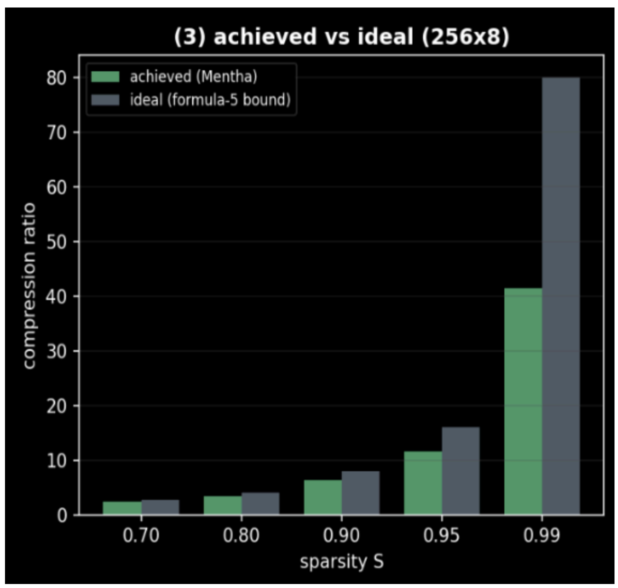
According to the above formula, we can obtain the compression ratio of matrix A as formula (5):

$$comRatio = \frac{M_b}{M_c} \leq \frac{1}{1-S} * \frac{dataWidth}{dataWidth + \log_2 M} \tag{5}$$

Sparsity	Our Implementation	Paper
0.70	2.4	2.4
0.80	3.5	3.5
0.90	6.5	6.6
0.95	12.2	12.3
0.99	47.1	47.9


(2) threshold cap (256x8)
thr=2
thr=3
thr=4
thr=8
thr=inf
compression ratio
sparsity S
0
5
10
15
20
25
30
35
40
0.70
0.75
0.80
0.85
0.90
0.95
1.00


(1) vs sparsity & block size
512x16
256x16
128x16
64x16
512x8
256x8
128x8
64x8
compression ratio (D*/D, storage)
sparsity S
0
10
20
30
40
0.70
0.75
0.80
0.85
0.90
0.95
1.00

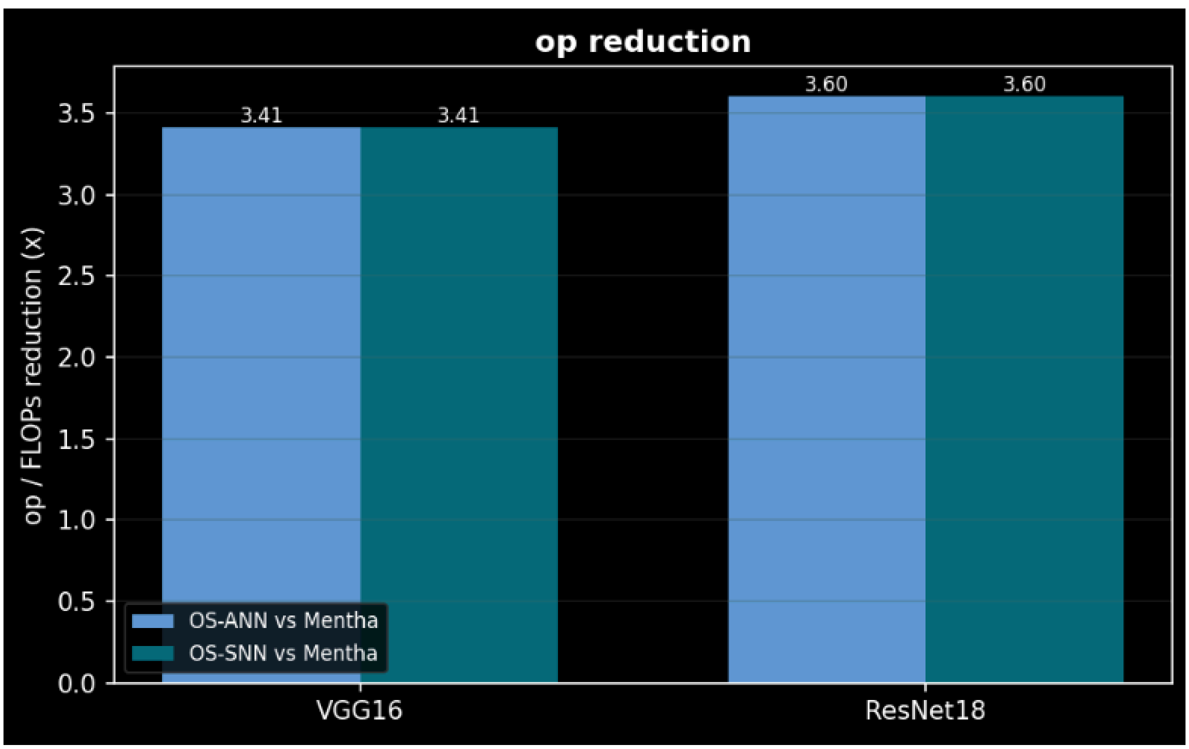

(3) achieved vs ideal (256x8)
achieved (Mentha)
ideal (formula-5 bound)
compression ratio
sparsity S
0
10
20
30
40
50
60
70
80
0.70
0.80
0.90
0.95
0.99

2.OS ANN/SNN vs OS Mentha ANN/SNN

Op Reduction: VGG16: 3.41x , ResNet18: 3.6x

$$\text{op reduction} = \frac{\text{dense MACs}}{\text{dual-side nonzero MACs}} = \frac{M \cdot K}{\sum_{m,k} \left[\text{act}_{m,k} \neq 0 \ \wedge \ \text{weight}_{m,k} \neq 0 \right]}$$

Normal dataflow vs Mentha

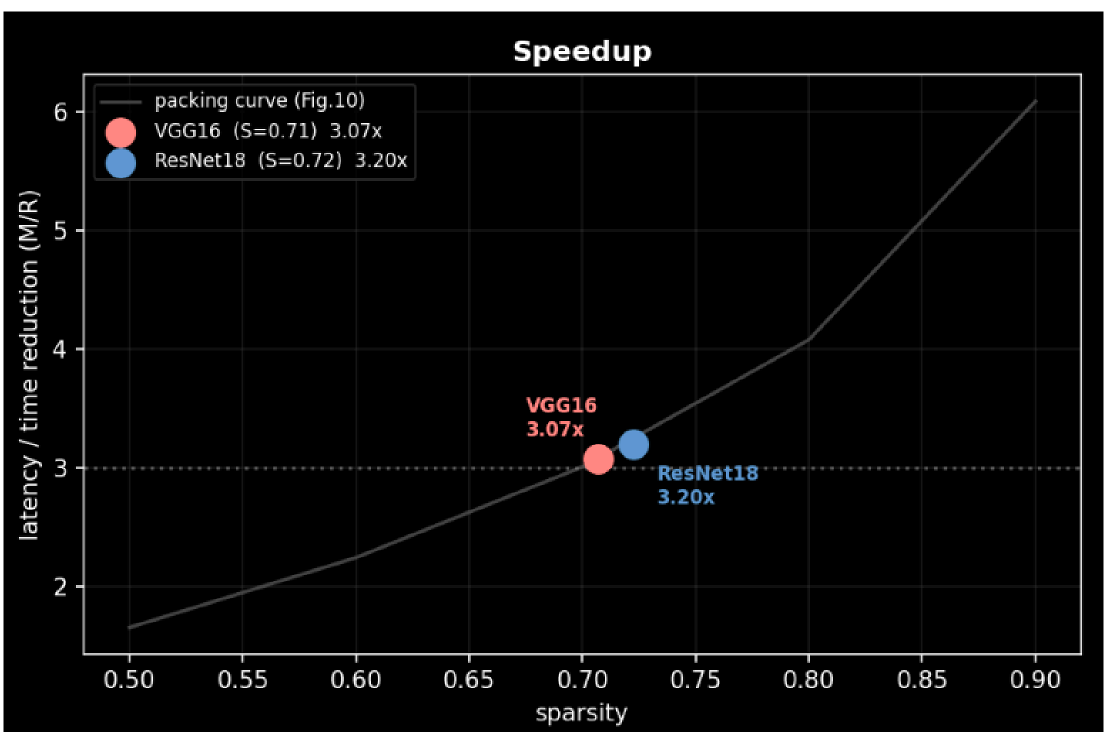


Speedup: VGG16: 3.07x , ResNet18: 3.20x

Speedup. The computation time is proportional to the amount of input data, and is also equivalent to the size of the input matrix. We define B as a $K \times N$-scale dense matrix. We can therefore determine that the calculating times for the baseline and Mentha are $T_{base} \propto M \cdot K \cdot N$ and $T_{comp} \propto R \cdot K \cdot N$ respectively. Based on formula (1),

we can obtain the upper limit of acceleration as shown in formula (6):

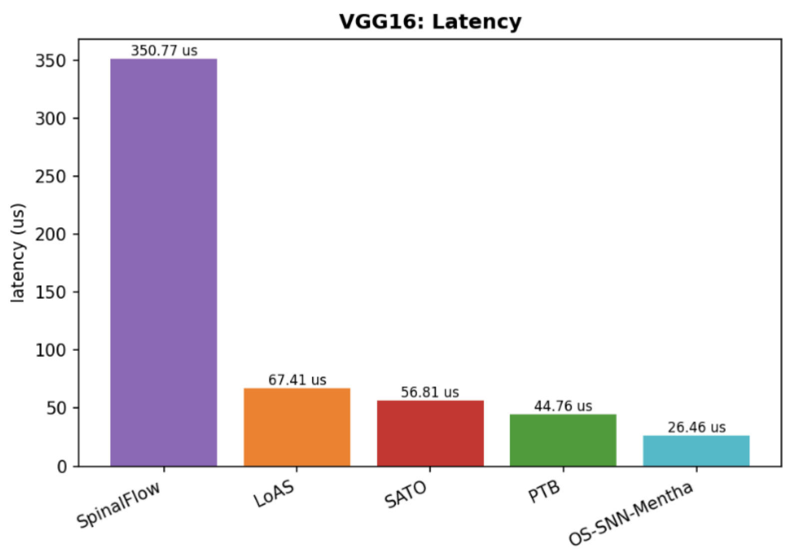
$$Speedup = \frac{T_{base}}{T_{comp}} \propto \frac{M}{R} \leq \frac{1}{1-S} \tag{6}$$



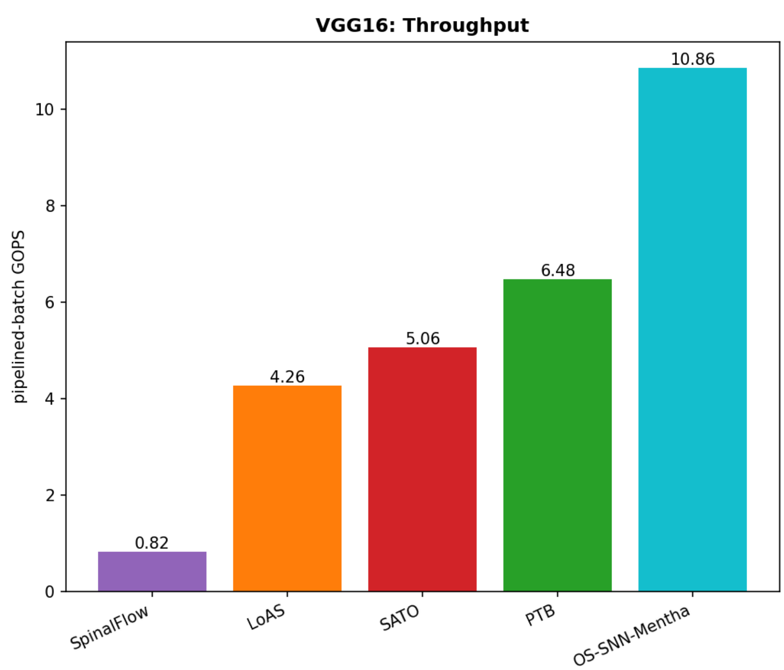
3. Baseline comparisons with PTB, SATO, LoAS, SpinalFlow

VGG16

Latency

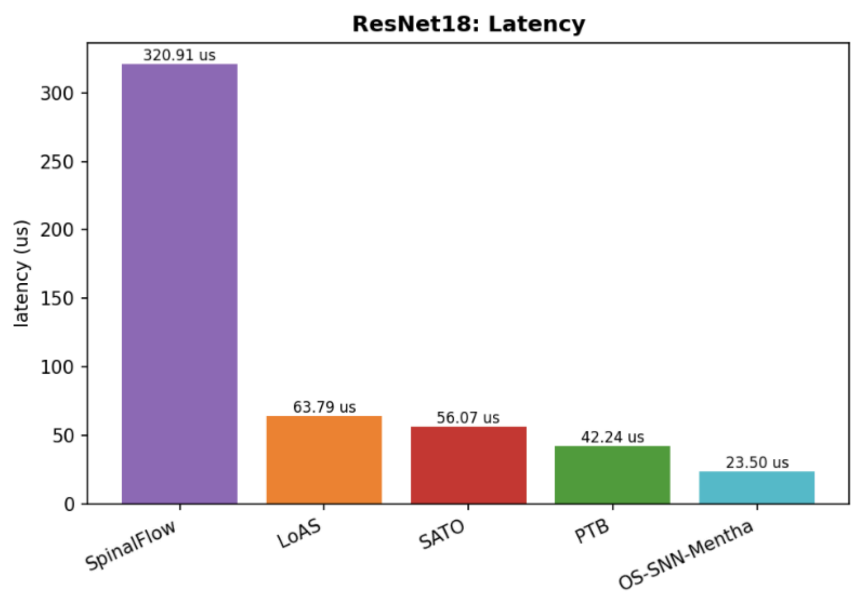


Throughput

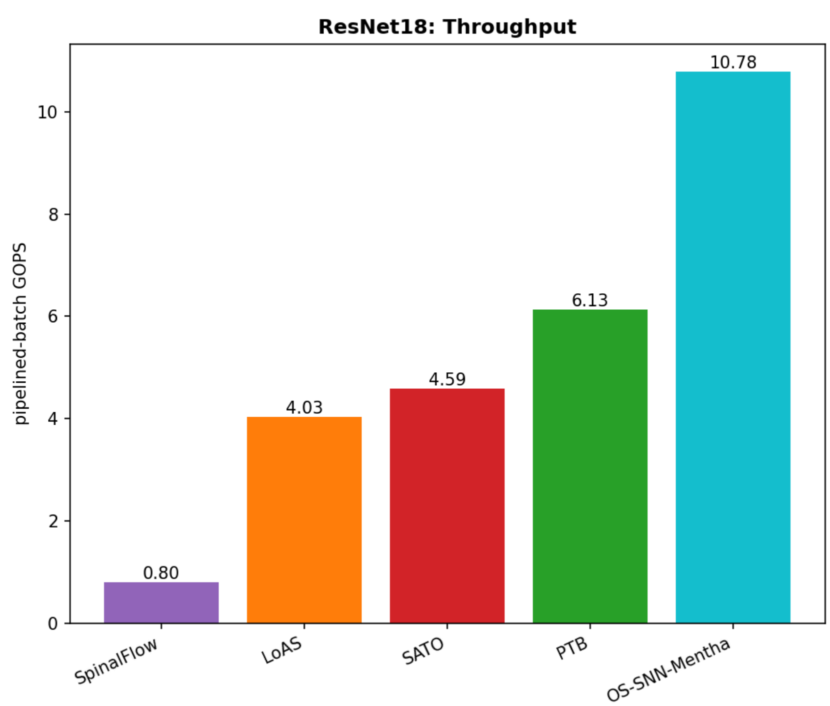


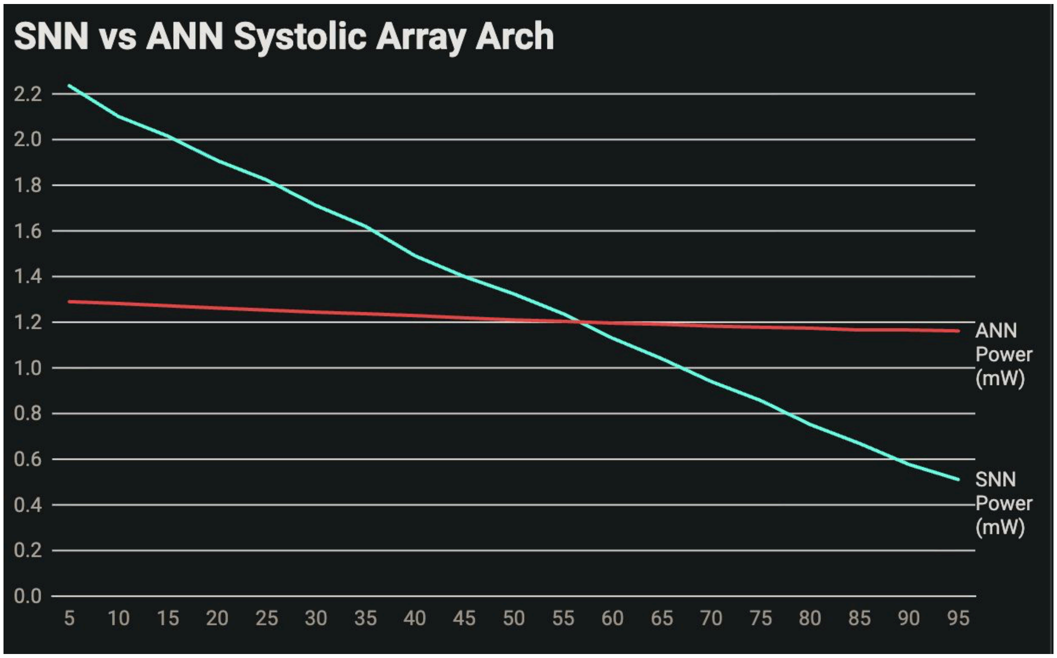
ResNet18

Latency



Throughput





Sparsity (%)	SNN Power (mW)	ANN Power (mW)
5	2.236	1.290
10	2.101	1.282
15	2.015	1.272
20	1.908	1.262
25	1.823	1.253
30	1.711	1.244
35	1.620	1.237
40	1.491	1.229
45	1.400	1.219
50	1.324	1.210
55	1.237	1.204
60	1.130	1.196
65	1.040	1.191
70	0.940	1.183
75	0.857	1.178
80	0.752	1.174
85	0.669	1.166
90	0.577	1.166
95	0.511	1.162

With packing: snn: 4.2 -> 2.9 , ann: 2.9 ->2.5